

ESO208 COMPUTATIONAL METHODS IN ENGINEERING

ASSIGNMENT 6

MAXIMUM MARKS: 30

DUE DATE: 27 SEPTEMBER 2026

Please upload hand-written solutions on Gradescope. Typed solutions except codes and plots will not be graded. Please upload single PDF file for hand-written part.

Q.1. [15 marks] The table below contains data of dissolved oxygen in sea water (mg/L) at different temperature levels.

$T(^{\circ}\text{C})$	0	8	16	24	32	40
O (mg/L)	14.621	11.843	9.870	8.418	7.305	6.413

- (a) Write a code to calculate the coefficients of a 5th order Newton's interpolating polynomial. [5]
- (b) Write the equations to fit quadratic spline to the given data (manually). [4]
- (c) Solve the set of equations in (b) using Gauss elimination (use code). [4]
- (d) Plot Newton's polynomial in (a) and quadratic spline in (c) on the same graph for $0 \leq T \leq 40$. Also show the data points in the table on the graph. [2]

Q.2. [15 marks] The following excel file contains annual flows in a river from 1906-2019: [assignment6.xlsx](#).

- (a) Calculate the discrete Fourier transform of the annual flow data by taking $t = 1$ for the year 1906 and $T = 114$ for the year 2019. First write the Fourier transform equation manually and then obtain the coefficients through code. (Please do not use in-built functions) [10].
- (b) Make a plot of power vs frequency using the results obtained in (a). [2].
- (c) Which are the three most dominant frequencies present in the flow time-series? [3]